WIRE DRAWING

Real-Time Pass Schedule Optimization: means through this model we can tell the industries workers …..that what parameters like number of passes , lubricant,diameter should we kept so that after certain number of passes the required strength of material we can get ….means in simple language u can say that ….assume u have mild steel material and from that u have to make ….. a wire of certain stength…then mechanically u will perform an experiment that initially u will kept no. Of passes 8 then 10….then may be 12 …or u can take lubricant initially mineral oil …but it doesnot give u desired property…then u have to change the wholw system….like number of passes and diameter…of wire…and lubricant ….so it will alternatively affect the cost of productiona and time …..but if indusstries use our model the initially by putting values they can ….judge what should be number of passes and die diameter and lubricant so that we can get desired property wire …..

Now talking about the output values through our model indivisually in what sense they are related to beeneficiation of industry…….

1. Exit Temp.—-

Once an all the input parameters are fed then…u will get an ouput parameter results…..so suppose u know at what temperatrue dioes lubricant begin to degrade….so after inputing parameters if u get temp ouput more then desired then u can change lubricant…..according to u r convience.

2. Drawing stress ——-

At the die exit the drawn wire is subjeccted to tensile forces or tensile drawing stresses….u initially know the wire material property …how much force it can sustain….the exit force u shoud kept in limit so tht the wire doesnot snap off means toot na jaye…..

3.Drawing force——

If force is too high then it can causes a thinning of cross sectioanl area …in short necking….and line breaks

If force is too high then it will directly affect the contact zone inside the dies ..jap cpstan wire ko bahut jayada force se pull krta hai…then the presseure between die and wire surface will incrase high presseure or localised contact die angle se stress badh skta hai…..jisse metallurgical and structural level pr problem ho skti hai….

4. Delamination risk ——-

Eliminates catastrophic failure and give good application for end users.